

in the same mental state. Indeed, we can consider that as something that does not facilitate the investigation of mental function, whose time is much faster. Some tasks are very difficult to maintain for 45 minutes.

When you deal with aspects of synaptic activity that you like to refer to mental processes that are faster than the ones that you want to investigate, you can rely on markers that primarily are related to changes in blood flow supply, whose activity is much faster.

Example: H_2^{15}O half-life is 2 minutes, so in 10-15 minutes the activity of oxygen is disappeared, so you can monitor changes in blood flow that occur in few seconds. In some cases, oxygenated blood reaches a peak in the brain, that is compatible with several neural functions. Changes in blood flow supply are tightly coupled in changes in glucose and oxygen consumption (synaptic activities), so it can be used to represent synaptic activity, but you use tracers with different metabolic fate time.

When I'm dealing with an asymmetric distribution of glucose consumption and I'd like to compare these, you need to have absolute values for each of the voxels. Once you know the amount of dose that you have injected and the body mass of the participant, then you are able to reconstruct the curve, interpolating the values. Therefore, you are able to express Deoxyglucose as mg/100g tissue/minute (generally the range of value is 10-15 mg) or ml/100g tissue/minute (generally 50-70 ml).

Once you have the absolute value, we have the parametric or dynamic image, and at that point we can do any statistical operation with these images. When you take several volumes, you are trying to do an interpolation, so that you can have images to overlap. An important step is the spatial normalization, so that each we can move our brain into a standard reference state and have the possibility to compare different realizations (for example from different people). Each point is characterized by 3 coordinates. We use ATLAS (space of coordinates of reference), the first was the Talairach, obtained cutting the brain in slices and for many years it was the main reference for all brain studies.

In functional neuroimaging studies, cognitive subtraction refers to an aspect of experimental design involving the comparison of two conditions or brain states that are presumed to differ in only one discrete feature (the independent variable). Pure insertion asserts that there are no interactions among the cognitive components of a task

Donders' subtractive Methods

The main approach that we got derives from experimental psychology, that grew up from subtraction approaches.

Donders' work attempts to describe the processes going on in the mind, by analyzing cognitive activity into separate stages (many scientists assumed that responding to a stimulus occurred instantaneously!)

Donders was particularly interested in "timing the mind" and used a subtraction technique to time the different mental processes that the brain goes through when faced with different tasks.

Donders performed experiments using reaction time tasks in 1868:

1. a simple reaction time task - e.g. you are seated in front of a panel that contains a light bulb and a response button. When the light comes on, you must press the button. 190 ms

2. a discrimination reaction time task - e.g. you are seated in front of a panel with 5 light bulbs and one response button. When the target light goes on you must press the button - but not if the 4 other lights come on. 240 ms

These 50 ms of difference is the time that the mental function needs to discriminate the target.

3. a choice reaction time task - e.g. you are seated in front of 5 light bulbs, each with their own button. You must press the button corresponding to the appropriate light. 280 ms

Together with discrimination you have to select the correct motor action. The idea is that as soon as you add pieces, your reaction time is longer and you can build up experiments that allow you to add piece, that you can subtract from another condition.

Example: For visual stimulation, eyes open and white light or control condition and eyes closed. The only difference is the perception of the white light. You subtract statistically the PET images, and you obtain a group average response, and the mean difference image is something that tells you that perception of the white light relies on changes in activity in the occipital cortex.

Example: Chess as a case of study. Why chess?

- Chess problems are complex.
- The ability of the chess player is measurable.
- There are multiple studies in which the cognitive psychology of chess is studied.

In this study they decided to split the chess game in 4 different mental stages that we could compare:

- B/W discrimination.

Control baseline condition. The idea was to define specifically what the brain is doing during the tasks. (Visual perception, discrimination of colors, reading, recalling motor command, decision, ocular movement, thumb movement).

- Spatial discrimination (is the closest piece to the X white?).

All the things we said before but with an additional condition, spatial analysis of distance. (Spatial attention, static spatial analysis). Subtracting the PET images, we are able to see the different areas of the brain working in the new task. (Precuneus, occipital lobe, superior parietal lobule, medium temporal gyrus, superior premotor cortex).

- Rules (can the white queen catch the black pawn).

You have to discriminate the value of the pieces and recall the rules, so you must activate the memory. (Value discrimination, rules recall, dynamic spatial analysis). The areas of the brain working are different (left hippocamp, left temporal lobe, left superior temporal gyrus, left medium temporal gyrus, left fusiform gyrus, right post-central gyrus, cerebellum).

- Checkmate (can the white player win in one move).

(Recall of the game rules for all the pieces, consequence valuation for every move, generation of moves sequence). This was a 1994 study.

Example: Four different experimental conditions, in which order we should make the patient do the different tasks.

- Visual
- Spatial
- Rules (memory)
- Strategy

The first thing we would say is to make him do them in this exact order, but we have to take in consideration the increase of difficulty for the different tasks. How can we discriminate between the areas of the brain that are working trying to solve a more difficult task (involving emotional harassment) and the ones specifically working for the task itself. How to manage task effect? They also learn to do the tasks, so the tasks become easier and brain response is somehow reduced, or you can become tired and habituated. We can't present them in casual order because we would have to compare different patients that had the tasks presented in different order.

You teach the patients the tasks, until they reach maybe 90% accuracy, and then you can have your experiment putting them in the machine. In this way you avoid training effect.

Face perception

Example (How to build a PET imaging experiment)

Which are the variables that we have to take into account to determine our patients?

- Gender (sexual orientation)
- Ethnicity (cross-race effect)
- Age
- Education
- Health status
- Drug-free
- Psychometric characterization
- Possible selection bias (mostly students)
- Sampling methodology
- Calculating sample size

Cross-race effect: when we are asked to recognize the identity of people that we are not familiar with (Chinese, afro Americans), our performance is decreased in respect to recognizing Caucasians for example.

Experimental paradigm (How can we define the stimuli?)

The stimuli allow to study what is the GESTALT, psychology that focuses on how humans naturally perceive patterns and organize sensory information into meaningful wholes, rather than just as collections of separate parts.